Computer Science Practical List 7

1. Write a menu driven program to implement a stack with the help of a list of integers. Use separate functions for performing PUSH and POP operations. POP operation should display a "Stack is EMPTY" message, if the list acting as a stack contains nothing in it. Call both the functions, with the help of menu driven options.

2. Given the following expected information of a list Book,
   - **Code of type integer**
   - **Title of the type string**

   Write a menu driven program to implement a stack of Book having two separate functions to perform PUSH and POP operations.

3. Write a Python code with the following functions in Python
   - PUSH(ITEMLIST,ST), where ITEMLIST is a nested list of lists containing INO (an integer), ITEM(a string), QTY(an integer). From this list, push all lists in Stack ST, which have QTY<10.
   - POP(ST), to remove and return an item from the top of STACK and should return a string with the content as "No reorder ITEM" when there are no more items left in ST to be popped.
   - DISPLAY(ST), to display the content of the stack if it has at least one element, otherwise display an appropriate error message as "No reorder ITEM".

   Also,
   - Initialize ITEMLIST with at least 5 items
   - Call PUSH function
   - Give menu driven options to the user to call POP and DISPLAY functions in a loop till the time POP does not return a message as "No reorder ITEM"

4. Write functions in Python (i) PUSH(N) to add new names (strings) as stack (ii) POP(N) to remove names from N as in stack and (iii) SHOW(N) to display content of N. Here N is a stack implemented by a list of names (strings).
   Call the above functions using menu driven options in a loop.

IMPORTANT: REFER TO MORE QUESTIONS GIVEN IN ASSIGNMENT BOOKLET

General Instructions:

i. Type the solution of all the problems using SQL statements using **Python** Language on **Python IDLE** or **Colab**

ii. Type in the following on top of each of your program as comment lines

```
# ------------------------------------------------------------#
# List-Program No     : L7-P1
# Developed By        : Reema Jones
# Date                : 17-November-2025    #
-------------------------------------------------------------#
```

iii. On successful execution with meaningful data, **copy** and **paste** the sample output at the bottom of the program as comment lines.

iv. **"Turn in"** the softcopy of the programs as a single document in response to this assignment submission with each program on separate pages. Once verified by the teacher, take out a hardcopy of **each program on separate pages** from the printer and get them signed by the respective computer teacher. Add all the printouts in the practical file and get the Index entry also signed.

v. Recommended Font, "**Courier New**", Style "**Bold**" Size "**10**" for all programs with single line spacing. Default indentation **2 Character** only.

```
# ------------------------------------------------------------#
# List-Program No : L7-P1
# Developed By    : PUJA DEY
# Date            : 17-November-2025
# ------------------------------------------------------------#

def PUSH(S):
    A = int(input("Data: "))
    S.append(A)

def POP(S):
    if S != []:
        print(S.pop())
    else:
        print("Stack is EMPTY")

def SHOW(S):
    print(S)

STACK = []

while True:
    print("P:PUSH  O:POP  S:SHOW  Q:QUIT")
    choice = input("Option: ")

    if choice == "P":
        PUSH(STACK)
    elif choice == "O":
        POP(STACK)
    elif choice == "S":
        SHOW(STACK)
    elif choice == "Q":
        break
    else:
        print("Invalid Option!!!")
```

```
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: P
Data: 45
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: P
Data: 30
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: P
Data: 12
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: O
12
```

```
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: O
30
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: O
45
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: O
Stack is EMPTY
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: S
[]
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: Q
```

```
# ------------------------------------------------------------#
# List-Program No : L7-P2
# Developed By    : PUJA DEY
# Date            : 17-November-2025
# ------------------------------------------------------------#

def PUSH(BOOK):
    C = int(input("Code: "))
    T = input("Title: ")
    BOOK.append([C, T])

def POP(BOOK):
    if BOOK != []:
        print(BOOK.pop())
    else:
        print("Stack is EMPTY")

def SHOW(BOOK):
    print(BOOK)

BOOK = []

while True:
    print("P:PUSH  O:POP  S:SHOW  Q:QUIT")
    ch = input("Option: ")

    if ch == "P":
        PUSH(BOOK)
    elif ch == "O":
        POP(BOOK)
    elif ch == "S":
        SHOW(BOOK)
    elif ch == "Q":
        break
    else:
        print("Invalid Option!!!")
```

```
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: P
Code: 101
Title: PYTHON
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: P
Code: 203
Title: AI BASICS
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: S
[[101, 'PYTHON'], [203, 'AI BASICS']]
```

```
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: O
[203, 'AI BASICS']
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: O
[101, 'PYTHON']
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: O
Stack is EMPTY
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: S
[]
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: Q
```

```
# ------------------------------------------------------------#
# List-Program No : L7-P3
# Developed By    : PUJA DEY
# Date            : 17-November-2025
# ------------------------------------------------------------#

def PUSH(ITEMLIST, ST):
    for I in ITEMLIST:
        if I[2] < 10:
            ST.append(I)

def POP(ST):
    if ST != []:
        return ST.pop()
    else:
        return "No reorder ITEM"

def DISPLAY(ST):
    if ST != []:
        print(ST)
    else:
        print("No reorder ITEM")

ITEMLIST = [
    [101, "PEN", 5],
    [102, "PENCIL", 12],
    [103, "ERASER", 3],
    [104, "SHEET", 25],
    [105, "MARKER", 7]
]

ST = []

PUSH(ITEMLIST, ST)

while True:
    print("O:POP  D:DISPLAY  Q:QUIT")
    option = input("Option: ")

    if option == "O":
        R = POP(ST)
        print(R)
        if R == "No reorder ITEM":
            break
    elif option == "D":
```

```
        DISPLAY(ST)
    elif option == "Q":
        break
    else:
        print("Invalid Option!!!")
```

```
O:POP  D:DISPLAY  Q:QUIT
Option: D
[[101, 'PEN', 5], [103, 'ERASER', 3], [105, 'MARKER', 7]]
O:POP  D:DISPLAY  Q:QUIT
Option: O
[105, 'MARKER', 7]
O:POP  D:DISPLAY  Q:QUIT
Option: O
[103, 'ERASER', 3]
O:POP  D:DISPLAY  Q:QUIT
Option: O
[101, 'PEN', 5]
O:POP  D:DISPLAY  Q:QUIT
Option: O
No reorder ITEM
```

```
# ------------------------------------------------------------#
# List-Program No : L7-P4
# Developed By    : PUJA DEY
# Date            : 17-November-2025
# ------------------------------------------------------------#

def PUSH(N):
    name = input("Name: ")
    N.append(name)

def POP(N):
    if N != []:
        print(N.pop())
    else:
        print("EMPTY STACK")

def SHOW(N):
    if N != []:
        print(N)
    else:
        print("EMPTY STACK")

N = []

while True:
    print("P:PUSH  O:POP  S:SHOW  Q:QUIT")
    ch = input("Option: ")

    if ch == "P":
        PUSH(N)
    elif ch == "O":
        POP(N)
    elif ch == "S":
        SHOW(N)
    elif ch == "Q":
        break
    else:
        print("Invalid Option!!!")
```

```
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: P
Name: APPLE
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: P
Name: MANGO
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: P
```

```
Name: PINEAPPLE
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: S
['APPLE', 'MANGO', 'PINEAPPLE']
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: O
PINEAPPLE
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: O
MANGO
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: O
APPLE
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: O
EMPTY STACK
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: S
EMPTY STACK
P:PUSH  O:POP  S:SHOW  Q:QUIT
Option: Q
```